Tirtawening Bandung

ChatWater Stages 1–4: a practical worked example

Water supply, wastewater collection and Bojongsoang wastewater treatment

Prepared October 30, 2026 | Reference water year: 2023 | English input guide

We use a real Indonesian utility that operates both water and wastewater services. We combine published utility information with clearly identified historical proxies and example assumptions to show how to complete the whole ChatWater workflow. [1–6]

This is a report-ready demonstration, not a verified Tirtawening emissions inventory, an approved investment, or evidence of utility endorsement.

How to use this guide

Step	What we enter or prepare
1 Assess	Utility profile; ECAM water and wastewater inputs; an additional climate-risk and water-safety worksheet.
2 Identify actions	A short list of justified actions, including custom wastewater and resilience actions.
3 Project plan	Eight planning fields, a 104-week schedule, an illustrative budget and measurable indicators.
4 Financing Proposal	All preliminary-concept fields, financing choices, an eight-section narrative and preparation gaps.

Our evidence labels

Label	Meaning
R | Reported	A figure or operating fact from an identified public source. We retain its year and boundary.
D | Derived	Our arithmetic using reported data. This does not make the result a utility-certified measurement.
P | Proxy	A historical value or a different statistical definition used temporarily for the demonstration.
A | Assumed	An illustrative number, target, cost, risk score or planning choice. It is not a reported utility fact.
Unknown	The verified input is unavailable. For actual reporting, we keep it unknown rather than treating it as zero.

Enter numbers without thousands separators. Use annual totals unless stated otherwise. For Unknown, use the ? control or leave the field blank; do not type the word into a numeric box. Bracketed source numbers refer to the linked references.

We inspected the live forms. CREAT and WHO climate-resilient Water Safety Plan modules are labeled “Not active yet.” We therefore supply their additional inputs as manual planning notes, not as completed in-app assessments. We do not claim that this Bandung case was run and validated end to end. [16]

The real utility and the limits of our evidence

Perumda Tirtawening Kota Bandung operates drinking-water and domestic wastewater services. Bojongsoang is in Kabupaten Bandung and serves wastewater from Kota Bandung. We keep the service area and the plant location distinct. Its treatment train uses anaerobic, facultative and maturation ponds, with inlet pumping, screening and grit removal. It is not an activated-sludge plant. [3, 4]

Published indicator	Value and reporting basis
Water customers; active domestic connections	169,373 total customers; 165,680 active domestic connections, 2023 [1]. These are different categories.
Population served; service-area population	965,283 served; 2,793,549 in the reported service area, 2023 [1].
Administrative population	2,569,107, 2023 [1]. Do not interchange this with the larger service-area denominator.
Water service coverage; supply hours	34.55%; 18 hours/day, 2023 [1]. Coverage uses the source's service-area denominator.
Non-revenue water (NRW)	39.42%, 2023 [1]. NRW includes unbilled authorized use as well as losses.
Production capacity; reservoir storage	2,789 L/s installed; 667 L/s unavailable; 28,400 m^3 reservoir capacity, 2023 [1].
Pipe length	126,300 m transmission; 1,487,522 m distribution, 2023 [1].
Water volume reported by BPS	36,622,383 m^3 "air disalurkan," 2023 [2]. It is not explicitly identified as ECAM billed consumption.
Tariff and cost recovery	Average tariff IDR 6,638/m^3; full-cost-recovery indicator 85.53%, 2023 [1].
Company financial context	Revenue IDR 331.032 billion; operating expense IDR 314.617 billion; net profit IDR 6.432 billion, 2023 [1].
Existing plans and staffing	RPAM exists; business plan 2022–2026; RISPAM 2021–2040; 700 employees in 2023 [1].

The 2024 performance book reports 2023 operations. We use this consistent, traceable water reference year, not a claim about the latest utility performance. The book reports an energy-cost indicator of IDR 79/m^3, not electricity consumption in kWh. It cannot substitute for meter readings. [1]

Our weakest evidence is annual wastewater and energy data: Bojongsoang's available detailed monthly dataset is from 2019. We do not relabel it as measured 2023 data. All investment amounts and future benefits below are assumptions.

Stage 1 | Utility profile and general settings

We open ChatWater in English, use a separate DEMO utility profile and create a new assessment. The utility profile can be shared by several assessments, so we check its association before editing an existing profile. [16]

Profile field	Our entry
Utility name	DEMO | Perumda Tirtawening Kota Bandung
City / regency, province	Kota Bandung, West Java, Indonesia; Bojongsoang WWTP is in Kabupaten Bandung.
Water supply; sewer network; central IPAL	We operate it, for all three [R: 3].
Household septic tanks and pits	Not our responsibility [A: individual household containment is outside this modeled operating boundary].
Desludging and faecal sludge treatment	Not sure / not answered. Desludging services are publicly described, but confirm the current LLTT/IPLT operating scope before selecting a combined responsibility category [3].
Capital budget for new investment	Partly; co-financing needed [A]. This is not an assessment of current borrowing capacity.
In-house engineering capacity	Some [A]. We assume internal operations staff plus specialist support.
Operational data	Partly measured, partly estimated [A: describes this example dataset].

General field	Our entry
Service type	Water and wastewater
Assessment name	DEMO Tirtawening Bandung | 2023 reference + historical proxies
Period	From 2023-01-01 to 2024-01-01. The end date is exclusive: 365 days.
Country; currency	Indonesia; IDR
Global warming potential	IPCC Fifth Assessment Report, without climate–carbon feedbacks: CH_4 = 28; N_2O = 265.
Grid emission factor	0.677 $kgCO_2e/kWh$ [A: retained application default for reproducibility, not a verified 2023 Java–Bali factor].

Paste into Comments

We use verified 2023 water indicators, historical 2019 Bojongsoang proxies and explicit test assumptions. Energy, nitrogen, sludge, discharge balance and emission-factor applicability are unverified. This is a worked example, not a measured inventory or an approved project. We exclude onsite sanitation, fuels, trucking and downstream sludge management. Zero bypass means an assumed test boundary, not observed zero pollution. Climate evidence from 2025–2026 informs forward planning only. See this guide's labeled input tables and sources.

Stage 1 | Water supply inputs

We add one substage each for Water abstraction, Water treatment and Water distribution. These aggregate the utility system for the example. We do not copy total utility electricity into each substage.

Field in ChatWater	Enter this value	Unit	Basis
Water supply: resident population	2793549	people	R [1]
Water abstraction: volume abstracted / conveyed	64000000	m^3/year	A
Water abstraction: grid energy consumed	1000000	kWh/year	A
Water treatment: volume treated	62000000	m^3/year	A
Water treatment: grid energy consumed	800000	kWh/year	A
Water distribution: serviced population	965283	people	R [1]
Water distribution: volume distributed	60452927	m^3/year	D + P
Water distribution: grid energy consumed	1200000	kWh/year	A
Water distribution: authorized consumption	37226912	m^3/year	D + A
Water distribution: billed consumption	36622383	m^3/year	P [2]
Grid factor in each water substage	0.677	$kgCO_2e$/kWh	A

Why these volumes are internally consistent

For the test only, we map the BPS volume to billed consumption. We then calculate distribution input = 36,622,383 ÷ (1 − 0.3942) = 60,452,927 m^3, rounded. We assume unbilled authorized use is 1% of distribution input, or 604,529 m^3. Authorized consumption becomes 36,622,383 + 604,529 = 37,226,912 m^3.

This reproduces 39.42% NRW by construction. Water loss is 38.42% because it excludes unbilled authorized use. Neither the distribution volume nor the match to NRW independently validates the BPS-to-billing mapping. Confirm system input and customer billing records before actual reporting.

Optional fields and switches

Choose Yes for water-efficiency inputs in distribution to expose authorized and billed consumption. If a treatment description is requested, enter “Aggregated conventional surface-water treatment plus other sources; verify plant-specific allocation.” Official sources include river water, springs and boreholes; do not describe all water as pumped groundwater. [3]

Keep pumping-efficiency and electromechanical-efficiency modules off until pump head, flow and electrical measurements are available. Keep stage operating costs unknown. For an optional training-only total water energy cost, enter IDR 4,500,000,000 = 3,000,000 kWh × assumed IDR 1,500/kWh. Do not treat company operating expense as the cost of one process.

Stage 1 | What the wastewater evidence supports

We add Wastewater collection and Wastewater treatment. We represent the connected sewer and Bojongsoang pond train, not every household in Bandung. We leave Onsite sanitation outside this run. Wastewater pumping stations include Gumuruh and Cijaura; Bojongsoang also has inlet screw pumps. The energy allocation must avoid counting any pump twice. [3, 4]

Item	What we use and what it means
Treatment technology [R]	Screening and grit removal, then anaerobic, facultative and maturation ponds in parallel sets A and B [4].
Historical design flow [R]	80,835 m^3/day [5]. This is a design reference, not current demonstrated treatment capacity.
Historical inlet flow [D]	24,566,522 m^3 in 2019, derived by multiplying each monthly average daily flow by calendar days and summing [5].
Historical outlet flow [D]	18,829,038 m^3 in 2019, calculated in the same way [5].
Historical flow discrepancy [D]	5,737,484 m^3 between inlet and outlet. The available records do not close this balance.
Historical influent BOD load [D]	2,250,840.14 kg/year, calculated from monthly inlet flows and monthly BOD results [5].
Historical effluent BOD load [D]	438,036.28 kg/year, calculated from monthly outlet flows and monthly BOD results [5].
What is missing	A verified 2023 sewer population, bypass record, energy allocation, nitrogen balance, sludge removals and representative pond emission factor.

Two corrections that matter

The 2019 flow table prints annual-average figures that do not agree with its monthly rows. We use the monthly rows and show the calculation in Appendix A. We do not copy the inconsistent printed average. [5]

Monthly BOD observations are not a continuous, flow-proportional sampling program. January and February results cover set A; later results cover A and B. Our annual load is a historical approximation, not a certified annual pollutant mass balance. [5]

Our explicit demonstration choice

We use the derived 2019 inlet volume and influent BOD as proxies. To create a simple closed-flow test, we assume the same annual volume is collected, treated and discharged, with no reuse or bypass in the modeled branch. This deliberately replaces the unexplained historical outlet balance. It does not establish that Bojongsoang actually has no bypass, water losses, withdrawals or reuse.

Stage 1 | Wastewater collection inputs

Field in ChatWater	Enter this value	Unit	Basis
Sanitation: resident population	2569107	people	R, 2023 [1]
Sanitation: wastewater generated	Unknown	m^3/year	No city total
Population connected to sewer network	500000	people	A
Volume of collected wastewater	24566522	m^3/year	D + P, 2019
Collected wastewater discharged untreated	0	m^3/year	A: test only
Collected wastewater sent to treatment	24566522	m^3/year	D + P + A
BOD_5 load in collected wastewater	2250840.14	kg/year	D + P, 2019
Total nitrogen in collected wastewater	491330.44	kg N/year	A: 20 mg/L
Grid electricity for collection	500000	kWh/year	A
Grid emission factor	0.677	$kgCO_2e$/kWh	A
CH_4 factor for sewer collection	0	$kgCH_4$/kgBOD	A: flowing
N_2O factor for sewer collection	0	kgN_2O-N/kgN	A
CH_4 factor for untreated discharge	0.021	$kgCH_4$/kgBOD	A, method
N_2O factor for untreated discharge	0.005	kgN_2O-N/kgN	A, method

We select flowing-sewer assumptions for the demonstration. Zero collection factors do not prove the real sewer is always aerobic or free of sediment, stagnation and emissions. Untreated-discharge factors have no numerical effect while the assumed untreated volume is zero. [7]

The 500,000-person value is an example denominator, not a verified 2023 customer count. It must not be used to claim current sewer coverage. The citywide sanitation-access percentage includes onsite services and is not Bojongsoang's sewer coverage. [6]

Extra settings

Keep pump-efficiency calculations off until measurements are available. Assign collection electricity to upstream sewer pumping and assign inlet screw pumps to treatment. Mark actual stage costs unknown. If a total wastewater energy-cost example is needed, use IDR 1,875,000,000 = 1,250,000 kWh × assumed IDR 1,500/kWh.

We do not estimate citywide wastewater generation by equating it to the sewer inlet flow. Infiltration, stormwater connections and non-sewered households make those quantities different. For an actual inventory, untreated discharge and sewer factors remain unknown until the operating records support them.

Stage 1 | Bojongsoang treatment inputs

Field in ChatWater	Enter this value	Unit	Basis
Serviced population	500000	people	A
Volume of treated wastewater	24566522	m^3/year	D + P
Discharged effluent to water body	24566522	m^3/year	A: closed flow
Influent BOD_5 load	2250840.14	kg/year	D + P
Effluent BOD_5 load	614163.05	kg/year	A: 25 mg/L
Influent total nitrogen load	491330.44	kg N/year	A: 20 mg/L
Effluent total nitrogen load	294798.26	kg N/year	A: 12 mg/L
Grid electricity consumed	750000	kWh/year	A
Grid emission factor	0.677	$kgCO_2e$/kWh	A
Sludge removed, dry weight	150000	kg dry/year	A
BOD_5 removed as sludge	150000	kg BOD/year	A: separate
CH_4 emission factor: treatment	0.48	$kgCH_4$/kgBOD	P: deep pond
N_2O emission factor: treatment	0	kgN_2O-N/kgN	P: anaerobic
CH_4 emission factor: discharge	0.021	$kgCH_4$/kgBOD	A: river type
N_2O emission factor: discharge	0.005	kgN_2O-N/kgN	A: Tier 1

The pollutant-load equation is volume (m^3) × concentration (mg/L) ÷ 1,000. Concentration and mass are different inputs. For example, 24,566,522 × 25 ÷ 1,000 = 614,163.05 kg BOD/year. Do not type “25” into the annual BOD-load field.

We enter sludge dry mass and sludge BOD separately. Their equal values are coincidental assumptions for this test, not a universal conversion. Pond sludge can remain stored for years; this removal quantity is not a measured annual sludge production rate.

For the test, select No for fuel engines, captured biogas, trucked reused water, avoided reuse emissions and sludge-management calculations. These switches mean “outside this run,” not verified absence at the utility. An anaerobic pond can emit methane even when no biogas is collected. Do not select “covered lagoon” merely to make treatment emissions zero.

Stage 1 | Factor choices and numerical checks

The pond train needs a better factor before actual reporting

For a reproducible screening run, we choose “Anaerobic deep lagoon (>2m depth)” for treatment CH_4. The historical study gives 3.5 m for anaerobic ponds, with shallower downstream ponds. Applying 0.48 to the entire aggregated train is a proxy, not a validated site factor. ECAM specifically advises additional or customized factors for treatment trains with multiple technologies. [5, 7]

We duplicate the assessment and test 0.12, the shallow/facultative pond default, to show sensitivity. Neither endpoint is a confidence bound or a complete model of the serial train. We do not average factors by pond area or multiply both factors by the full influent load. A verified mass balance and suitable multi-stage method are required.

The treatment N_2O default of zero is an anaerobic-pond screening assumption. It does not demonstrate zero N_2O across facultative and maturation ponds. Likewise, 0.005 for discharge is a generic Tier 1 factor; assess the receiving water before substituting a site-specific or nutrient-impacted factor. [7]

Check	Expected demonstration result
Water balance	Billed ≤ authorized ≤ distributed ≤ treated ≤ abstracted.
NRW; water loss	39.42% NRW; 38.42% water loss. Rounded-volume differences are negligible.
Water electricity	3,000,000 kWh; 2,031 tCO_2e at factor 0.677.
Wastewater electricity	1,250,000 kWh; 846.25 tCO_2e at factor 0.677.
All included electricity	4,250,000 kWh; 2,877.25 tCO_2e. This is an assumed scenario, not reported utility emissions.
WWTP energy intensity	0.0305 kWh/m^3 treated. This low assumed value is for ponds and pumping, not activated-sludge aeration.
BOD load removal	1,636,677.09 kg/year, or approximately 72.71% of influent load, under the test assumptions.

ECAM treatment CH_4 uses (influent BOD − BOD removed as sludge) × treatment factor × CH_4 GWP. It does not additionally subtract effluent BOD in that equation. Discharged-water emissions use effluent loads separately. Follow the implemented methodology rather than replacing its formula with a different mass-balance equation. [7, equations 80 and 99]

We confirm values only after checking their intended use. An app “confirmed” flag is not scientific verification. Unknown values can leave totals incomplete. We do not use an aggregate emissions total from this exercise as Tirtawening’s inventory or claim any modeled reduction as achieved.

Stage 1 | Additional climate-risk inputs

CREAT is not active in this version. We retain this worksheet with our assessment and paste its key findings into Stage 3 and Stage 4. We use an asset–hazard–consequence–action structure inspired by CREAT, without claiming to have run the EPA tool. [8, 16]

Input	Our example entry
Planning horizon	2027–2040 [A]. The investment lasts 104 weeks after funding, with longer-lived assets tested beyond completion.
Assets and service dependencies	Cikalong raw-water intake; source-to-treatment supply; selected distribution zones; Gumuruh/Cijaura pumping; Bojongsoang inlet, power equipment and ponds [3, 4]. Exact elevations and boundaries remain to be surveyed.
Observed dry-season evidence	Tirtawening's September 2, 2026 notice reports lower Cikalong raw-water flow and distribution disruption [9]. It does not quantify a climate-change trend.
Observed flood evidence	Local government reported flooding and a levee breach in Desa Bojongsoang in February 2025 [10]. This establishes local hazard context, not proven inundation of the plant.
Projection information to obtain	BMKG rainfall projections and local observations for source catchments and assets. BMKG's portal uses CMIP6 scenarios and 25 km data; local hydraulic design needs finer analysis [11].
Scenario choice	Review SSP2-4.5 and SSP5-8.5, near-term 2021–2050 relative to 1981–2010, plus observed extremes [A selection using the BMKG framework]. No local percentage change has been extracted here.
Dry-season stress test	Assume available source yield is 20% lower for 30 days [A, not a forecast]. Calculate storage, demand restrictions and priority-service needs from actual operating data.
Wet-weather stress test	Assume peak sewer inflow reaches 1.5 times measured dry-weather peak, and a 6-hour power outage occurs [A]. Use measured peak flow, not annual average, for design.
Flood design input	Unknown. Survey ground, equipment and water levels; obtain flood depths, duration and return-period estimates. Do not use a news-reported road depth as plant design level.

We separate three statements: an observed service disruption, a plausible climate-related vulnerability, and a quantified future projection. Only the first two are supported here. Climate attribution and local future hazard magnitudes remain preparation tasks.

Stage 1 | Water safety and risk priorities

Tirtawening already reported an RPAM. We propose updating it with climate considerations, not creating a plan on the assumption none exists. WHO guidance supports source-to-consumer risk management and climate resilience. [1, 12]

Hazard and consequence	Initial priority [A]	Controls and evidence needed
Dry-season low source flow reduces supply continuity.	Likelihood 4 × consequence 4 = 16, high	Update drought triggers, source monitoring and demand management; verify available storage and alternative supply quality.
Heavy rain changes raw-water quality and may overwhelm treatment control.	3 × 4 = 12, high	Track turbidity, treatment performance and disinfectant residual; establish operating limits and shutdown/escalation rules.
Intermittent or low-pressure supply permits contamination entry through leaks.	3 × 5 = 15, high	Pressure logging, leakage repair, safe restart/flushing and water-quality verification. Do not reduce pressure below a validated safe operating range.
Flooding or power interruption disables sewage conveyance or inlet pumping.	3 × 5 = 15, high	Survey electrical elevations; test safe backup arrangements, alarms, maintenance access and overflow response.
Pond overloading, accumulated sludge or short-circuiting worsens effluent quality.	3 × 4 = 12, high	Measure flow and loads; inspect hydraulics; survey sludge depth; check actual capacity and discharge requirements before expansion.

Our workshop scale is 1–5 for likelihood and consequence; scores 1–4 are low, 5–9 medium, 10–16 high and 17–25 very high. These are illustrative team judgments, not official CREAT outputs, observed probabilities or financial loss estimates.

Water-safety worksheet to retain with the project

We record the hazard, hazardous event, affected users, existing control, control limit, monitoring frequency, responsible role, corrective response and verification record. Utility staff must set actual operating limits from the approved RPAM, treatment design and current requirements.

Example entry: “Low source flow during a dry period reduces available supply. We monitor source flow daily during an alert, coordinate distribution and communicate service changes. The operations manager triggers the approved drought response when the locally validated threshold is reached. We record supply hours, pressure, residual disinfectant and complaints, with special attention to health facilities and vulnerable households.”

We do not invent a universal chlorine, turbidity, flood-elevation or pressure limit. Those are engineering and public-health control values that require the utility’s system data and applicable standards.

Stage 2 | Choose actions with reasons

After Stage 1, we open View results and continue to action identification. We check that the assessment name is Bandung. In our form review, an earlier assessment's actions remained visible while the new assessment was incomplete. We do not reuse another case's action totals or plan. [16]

Action	Our choice and reason
Reduce water loss	Select, then narrow the scope to three demonstration DMAs. Published NRW is 39.42% [1]. Confirm physical versus apparent losses and service pressure before estimating benefits.
Existing northern-area NRW contract	Coordinate before selecting sites. The utility describes a long-term Adaro performance-based contract with a target of 17% NRW from a 2020 northern-area baseline of 45.79% [13]. Do not duplicate contract areas, funding or credited savings.
Verify pump performance	Select diagnostics for relevant water and wastewater pumps. We lack measured head, flow and input power. Testing is justified; immediate replacement is not yet justified.
Bojongsoang reliability and climate risk	Add a custom action. Metering, pond condition, power resilience and flood exposure need verification before expansion or methane investment.
Climate-resilient RPAM update	Add a custom action. Link drought, wet-weather quality and distribution continuity to the existing plan.
400 additional sewer connections	Add as a conditional phase. Proceed only after capacity, effluent performance, network levels, affordability and permissions are verified.
Solar PV	Defer procurement. Confirm load profiles, site/roof conditions, ownership, grid arrangements and lifecycle cost. A generic percentage saving is not a site design.
Aeration optimization	Do not select an activated-sludge aeration action for Bojongsoang's pond train. Inlet pumping and pond performance are the relevant starting points.
Biogas capture / covered ponds	Select feasibility investigation only within the custom action. No construction or methane-reduction claim before gas yield, hydraulics, structural design and safety are assessed.
Route untreated sewage to treatment	Do not infer a real bypass problem or its absence from our assumed zero. Add this action only when measured overflow/bypass evidence and spare treatment capacity support it.

Action identifiers and automatic suggestions can change. We select by action title and location, review default savings assumptions, and use Save selected actions. For custom actions, ChatWater states that benefits are not calculated or scientifically reviewed. [16]

Stage 2 | Copy-ready custom actions

Use “+ Add your own action.” Use each heading after “Action B1/B2/B3:” as the Title, then enter the four fields below. The entries fit the displayed character limits. [16]

Action B1: Bojongsoang reliability and climate-risk assessment

Field	Paste this entry
What will be done	We will verify inlet/outlet meters and pump energy, sample BOD and total nitrogen, survey pond sludge and hydraulic condition, map flood exposure and test emergency power arrangements. We will prepare a prioritized rehabilitation design and a separate methane-capture feasibility decision.
Problem addressed	Historical flow records do not close the water balance. Current loads, energy, bypasses and pond performance require verification before expansion or emissions-reduction claims.
Location	Wastewater treatment › Wastewater treatment 1
Notes	We model the actual anaerobic–facultative–maturation pond train. We do not assume activated sludge, zero fugitive methane or safe agricultural reuse. Coordinate with Kabupaten Bandung and relevant river and environmental authorities. Methane works are outside the initial construction budget.

Action B2: Climate-resilient RPAM and service-continuity update

Field	Paste this entry
What will be done	We will update the existing RPAM with dry-season source-yield and wet-weather water-quality risks, pressure and continuity monitoring, emergency communication, priority-user support and staff drills. We will maintain a linked wastewater flood and power-loss response procedure.
Problem addressed	Recent source-flow disruption and local flood exposure create plausible service risks. Asset-specific future climate and consequence data are incomplete.
Location	Whole assessment
Notes	Use local observations and BMKG projections. The 20% yield reduction and 6-hour outage are stress tests, not forecasts. Utility staff will approve operating triggers and validate responsibilities.

Action B3: Conditional pilot of 400 sewer connections

Field	Paste this entry
What will be done	We will survey an eligible sewered catchment and develop 400 proposed connections. Construction proceeds only after treatment-capacity and effluent checks, network design, consent, affordability support and funding approval.
Problem addressed	Some households may remain unconnected even where a feasible sewer connection exists. The eligible sites and actual beneficiaries are not yet confirmed.
Location	Wastewater collection › Wastewater collection 1
Notes	We allow IDR 3.2 billion using the historical 2024 grant amount per connection as a rough reference, not a current quotation. If capacity or safeguard checks fail, defer this phase and rescope the budget; do not force new inflow into an overloaded system.

Stage 3 | Project summary and fields 1–4

Project summary

We propose a 104-week Tirtawening water-loss and wastewater-reliability pilot. We will establish a measured baseline, work in three water DMAs outside or formally coordinated with the existing northern-area NRW contract, assess and improve selected sewer/WWTP reliability controls, update the RPAM for climate risks, and conditionally connect 400 households. The illustrative investment envelope is IDR 10 billion. Sites, engineering quantities, financing and all benefit estimates remain subject to verification.

1. Define the project

Our objective is to improve dry-season water-service reliability and wet-weather wastewater-service continuity while reducing avoidable losses and energy use. Our demonstration scope is three DMAs with approximately 3,000 water connections, selected pump and control assets, Bojongsoang diagnostics and limited rehabilitation, and up to 400 new sewer connections. All scope quantities are assumptions. We exclude a new treatment plant, complete pond reconstruction and methane-capture construction from this initial budget.

2. Establish the baseline

We retain the published 2023 water context: 39.42% NRW, 18 supply hours/day and 169,373 customers. These company indicators are not the measured baseline for the chosen pilot zones. We collect at least eight weeks of initial DMA flow, pressure, billing and pump data, supported by available 12-month records. We reconcile Bojongsoang inflow/outflow, bypasses, BOD, nitrogen, electricity and sludge records. The 2019 wastewater proxy and assumed electricity inputs are replaced before claiming project benefits. We document wet/dry-season differences and data quality. If adequate historical records are unavailable, we extend baseline monitoring before fixing benefits or construction scope.

3. Identify project activities

We confirm the contract and asset boundary; calibrate meters; undertake leak detection, repairs and appropriate pressure management; verify pump duty and efficiency; inspect ponds and sludge; complete flood and power-risk screening; update operating and emergency procedures; prepare designs and procurement; execute justified limited works; and monitor performance. New sewer connections are a conditional work package. Methane capture remains a feasibility question, not a guaranteed mitigation activity.

4. Assign responsibilities

We propose Tirtawening’s designated project manager as accountable lead, supported by water-distribution, wastewater-operations, engineering, laboratory, finance and procurement functions. A specialist consultant supports hydraulic, climate, environmental and economic analysis. Kota Bandung supports public coordination and financing discussions; Kabupaten Bandung and relevant authorities are consulted for works affecting the plant location and receiving waters. We confirm actual mandates, named staff, land rights and approvals before implementation. No organization has committed through this example.

Stage 3 | Fields 5–8

5. Develop the implementation schedule

We plan 104 weeks from confirmed funding and authorization. Weeks 1–8 establish data and responsibilities; weeks 9–16 complete feasibility and design; weeks 17–24 finalize approvals and financing; weeks 25–36 cover procurement; weeks 37–52 cover works and commissioning; and weeks 53–104 verify performance across a full annual cycle. Monitoring begins during baseline development. Procurement and works remain gated by the approved design, environmental requirements and funding. We do not assign a calendar start date before these conditions are met.

6. Define technical and operational requirements

We require calibrated bulk and customer meters, timestamped flow/pressure and electricity data, pump duty-point measurements, verified network maps, laboratory quality assurance, pond/sludge surveys and safe access. Design must preserve drinking-water quality and minimum operating pressure, maintain sewage conveyance during construction and demonstrate treatment capacity before new connections. We include spare parts, staff training, maintenance responsibilities, emergency power arrangements where justified, cybersecurity/access controls for telemetry, and a funded operating budget. Equipment quantities and specifications follow the survey, not this example.

7. Assess implementation risks

Our main risks are overlap with the existing NRW contract; inaccurate meters; mistaken estimates of real leakage; dry-season interruption; flood exposure or power loss; pond overloading; construction disruption; land or access constraints; unaffordable household connections; procurement delay; and unfunded maintenance. We manage these through a contract-boundary review, calibrated baseline, staged works, safe service continuity procedures, early community engagement, affordability assessment and capacity/effluent decision gates. We record residual risk and an accountable owner before procurement.

8. Establish monitoring and evaluation

We measure DMA input, billed and unbilled authorized use separately, pressure, supply hours, complaints and repair performance. We compare pump electricity at equivalent hydraulic duty and service output. For wastewater we track metered inflow/outflow, bypass events, BOD and nitrogen loads, effluent results, pond condition and functional new connections. We verify training and emergency exercises. We report baseline, target, result, source, uncertainty and responsible role, with separate water and wastewater beneficiary counts and checks for overlap. We do not claim emissions savings until the baseline, factors and post-project data are accepted.

Review before saving

We use Review details, replace generic AI text with these case-specific entries, and resolve or document each missing-data item. A saved draft is not evidence of feasibility, utility approval or funding eligibility. We save the project plan only under the Bandung assessment. [16]

Stage 3 | Schedule and decision gates

ID / task	Owner role [A]	Start / weeks	After
T1 Baseline, survey and boundary review	Utility project manager + operations	1 / 8	None
T2 Feasibility, risk assessment and design	Engineering + specialist consultant	9 / 8	T1
T3 Approvals, safeguards and financing	Project manager + finance	17 / 8	T2
T4 Procurement	Procurement + engineering	25 / 12	T3
T5 Works, training and commissioning	Contractor + utility operations	37 / 16	T4
T6 Performance monitoring and evaluation	Operations + laboratory + finance	53 / 52	T5

These are sequential planning allowances, not commitments by the utility or a contractor. We enter Weeks and After in the task table; start weeks should follow the dependency logic. Activities such as stakeholder engagement and routine monitoring run throughout.

Gate	Evidence needed to proceed
G1: confirm pilot scope, end of week 8	Mapped DMAs and customer list; written check against the Adaro contract; preliminary wastewater flow balance and asset responsibilities.
G2: approve design, end of week 16	Verified leakage and hydraulic diagnosis; pump duty tests; pond and flood assessment; cost estimate; current environmental and operating requirements.
G3: commit procurement, end of week 24	Approved funding, procurement plan, land/access arrangements, safeguards and operating-cost provision.
G4: authorize 400 connections	Capacity and effluent evidence, network levels, household eligibility/consent, affordability measures and approved design. Failure defers this package.
G5: accept works, end of week 52	Commissioning tests; as-built records; operator training; spare parts; water-quality and wastewater-performance checks.
G6: verify benefits, end of week 104	Twelve months of comparable post-commissioning records, seasonal adjustments, independent review where required and documented uncertainty.

We do not infer spare capacity from annual average flow alone. The historical average inlet of about 67,306 m^3/day is below the design reference, while individual historical monthly averages exceed it. Current peak flows, organic loads, pond condition and effluent results determine whether expansion is feasible. [5]

Stage 3 | Illustrative investment budget

All figures below are assumptions in IDR, for concept-stage discussion. They are rounded allowances, not procurement estimates or commitments. We use a gross funding envelope; detailed tax treatment, escalation and procurement quantities remain to be priced.

Work package	Basis [A unless stated]	IDR billion
Baseline, surveys, engineering and safeguards	One integrated preparation package	0.80
Three DMA metering, telemetry and leak-repair pilots	3 × IDR 0.80 billion allowance	2.40
Pump tests, metering and justified minor rehabilitation	Selected water/sewer assets; quantities to be designed	0.80
Bojongsoang monitoring and reliability controls	Limited metering, controls and justified flood/power measures	1.20
Conditional 400 sewer connections	400 × IDR 8 million historical grant reference	3.20
RPAM update, training, engagement and monitoring	One integrated support package	0.60
Base estimate	Sum of six packages	9.00
Contingency	IDR 1 billion, approximately 11.1% of base	1.00
Total illustrative investment	Capital and project-preparation envelope	10.00

Bandung's 2024 performance report records 577 wastewater connections and IDR 4.616 billion under Hibah Air Limbah. Dividing gives IDR 8 million per connection. We use this only as a historical grant-per-connection reference. It is not a verified current construction unit cost, a current open funding program or a guarantee that all 400 connections cost the same. [6]

Operating cost and financing assumptions

We allow an additional IDR 300 million/year after commissioning for incremental maintenance, monitoring, calibration and support. This is separate from the IDR 10 billion investment and must be refined and funded. It excludes any large increase in pumping, treatment, sludge removal or major periodic replacement costs.

Example financing mix: IDR 2 billion utility contribution, IDR 3 billion local/national public support and IDR 5 billion external development or climate support. All three are uncommitted assumptions. If any source is unavailable, we phase or redesign the project before procurement.

The IDR 10 billion envelope is approximately 3.0% of reported 2023 company revenue. That comparison does not establish affordability: cash flow, debt obligations, tariff constraints, connection charges and ongoing operating costs still require review. [1]

Stage 3 | Targets, benefits and verification

Indicator	Example target [A]	How we verify
DMA scope	3 DMAs, approximately 3,000 water connections	Mapped boundaries and active customer register; exclude overlap with the existing contract.
Pilot NRW	39.42% to 34.42% after 12 months	Initial 39.42% is borrowed from company context, not measured pilot NRW. Replace with the pilot baseline.
Recovered water, conditional illustration	90,000 m^3/year at fixed pilot input of 1.8 million m^3/year	1.8 million × 5 percentage points. Separate apparent-loss recovery from actual physical water savings.
Service continuity	Pilot average 18 to at least 20 hours/day	18 hours is a provisional borrowed baseline. Use customer/pressure records and protect water quality.
Pump energy	Target 10% reduction in tested pump kWh at comparable duty	Measured head, volume, run hours and power; target is conditional on diagnostics, not guaranteed.
Wastewater reliability	100% of target meters calibrated; all events logged	Calibration certificates, complete flow/energy records and overflow/outage logs.
New sewer connections	Up to 400 functional connections	Commissioning, active service and household count; only after the capacity gate.
Water safety and emergency response	Updated RPAM; two documented drills	Approved procedures, attendance, response records and corrective actions.
Effluent quality	Meet applicable permit/approval requirements	Representative sampling and laboratory results; no unsupported claim of current compliance.
GHG outcome	No fixed tCO_2e target at this stage	Recalculate using verified activity data, appropriate factors and consistent boundaries.

For an optional energy illustration, 100,000 kWh/year of verified savings would equal 67.7 tCO_2e/year at the assumed factor 0.677 and IDR 150 million/year at assumed IDR 1,500/kWh. We do not add this to other pump or leakage savings unless we establish they are independent.

Recovered water can improve service instead of reducing production. In that case, it may deliver adaptation and service benefits without an equal reduction in energy. A billing recovery is a revenue benefit, not necessarily a water or GHG saving. We do not value all recovered water at the average tariff and call the result net profit.

Stage 4 | Preliminary-concept fields 1–7

The active route is the preliminary GCF concept. Other financing routes are shown as planned in the reviewed version. We can prepare a concept for discussion, but the app is not a funding application or approval system. [16]

Field	Our copy-ready entry
Project title	Bandung Water-Loss and Wastewater Reliability Pilot
Location	Kota Bandung, West Java, Indonesia, with wastewater-treatment works at Bojongsoang in Kabupaten Bandung. Exact DMA, sewer and asset boundaries will be confirmed during preparation.
Organization	Perumda Tirtawening Kota Bandung is the illustrative implementing utility. This is a public-data worked example; utility participation and submission authority have not been confirmed.
Objective	We aim to improve dry-season water-service continuity and wet-weather wastewater reliability, reduce avoidable physical losses and energy use, and build a measured basis for future low-emission wastewater investment.
Climate problem	Lower raw-water flow at Cikalong disrupted service during the 2026 dry season. Local flooding has occurred around Bojongsoang. These observations identify vulnerabilities, but asset-specific projections, flood levels and future service consequences still need study. We will use local observations and BMKG scenarios to establish the climate rationale.
Investment need	We propose an illustrative IDR 10 billion package for metering, three DMA pilots, pump diagnostics and limited rehabilitation, Bojongsoang reliability controls, RPAM updating and up to 400 conditional sewer connections. Major pond reconstruction and methane-capture construction are excluded.
Beneficiaries / service area	Our assumed direct scope is approximately 3,000 existing water connections in three DMAs and up to 400 new sewer connections. We will verify households and people, vulnerable groups and overlap. The utility-wide figure of 965,283 people served is context, not the pilot’s beneficiary count.

If a people-count field is required by a later template, we first obtain actual household/connection occupancy and avoid counting customers receiving both services twice. We do not assume every utility customer benefits directly from a small pilot.

Stage 4 | Preliminary-concept fields 8–14

Field	Our copy-ready entry
Purpose	Mitigation and adaptation. The mitigation potential remains unquantified until the measured baseline and feasible designs are available.
Adaptation evidence	We cite Tirtawening's September 2, 2026 Cikalong source-flow notice and the February 2025 Bojongsoang flood report as observations. We will add catchment and asset projections, flood and drought analysis, service consequences and vulnerability data. A 20% source-yield reduction and 6-hour outage are illustrative stress tests, not forecasts.
Duration	104 weeks from confirmed funding and authorization; approximately 24 months. The first year covers preparation, procurement and works; the second year verifies operating performance.
Total cost	IDR 10,000,000,000 illustrative investment, including IDR 1,000,000,000 contingency. Additional operating-cost allowance: IDR 300,000,000 per year after commissioning, subject to detailed costing.
Financing source	Illustrative mix: IDR 2 billion utility funds, IDR 3 billion local/national public support and IDR 5 billion external support. None is committed. We first examine ordinary utility and public funding; a GCF route would require a suitable accredited partner and a stronger program-level climate case.
Partners	Proposed roles only: Tirtawening, Kota Bandung, Kabupaten Bandung where relevant, water-resources and environmental authorities, BMKG/data providers, communities and specialist advisers. PT Sarana Multi Infrastruktur is a possible GCF accredited-entity discussion partner, not a committed financier.
Assumptions	All sites, scope quantities, costs, targets and funding shares are provisional. We must resolve existing-contract overlap, current wastewater capacity, environmental requirements, land/access, household affordability, O&M funding and emissions-factor applicability. Historical wastewater proxies and assumed energy values cannot support a final funding or GHG claim.

We choose Create preliminary concept only after checking the selected assessment and the saved project plan. We review the generated narrative against these entries, remove invented local facts and retain every material preparation gap. We do not describe the result as approved, submitted, bankable or submission-ready. [16]

Stage 4 | Eight-section concept narrative

Use the following eight short sections to review or replace the generated discussion draft. These are drafting sections, not a claim that ChatWater reproduces every field in the current official GCF template.

1. Project and applicant

We present a 104-week, IDR 10 billion pilot for Tirtawening's water and wastewater services. Kota Bandung is the service location; Bojongsoang treatment works are in Kabupaten Bandung. This example does not establish application authority or partner commitment.

2. Climate rationale

Observed source-flow disruption and nearby flooding justify investigating water-service and wastewater-system vulnerability. We will quantify asset exposure, future hazards and service consequences using local observations, appropriate projections and engineering analysis. Routine repairs alone do not establish climate additionality.

3. Activities and outputs

We will verify data, implement three DMA pilots, test pumps, improve justified reliability controls, update the RPAM and conditionally add 400 sewer connections. Outputs include calibrated meters, validated designs, trained staff and a year of comparable monitoring records.

4. Expected outcomes

We aim for better service continuity, lower physical leakage, more reliable sewage conveyance and treatment, and evidence-based energy reduction. The proposed NRW, energy and connection targets are assumptions requiring measured baselines. We make no verified emissions-reduction claim.

5. Implementation arrangements

A designated utility project manager coordinates operations, engineering, laboratory, finance and procurement. Government, community and specialist roles will be agreed during preparation. Investment gates prevent duplicate funding and connections beyond demonstrated treatment capacity.

6. Budget and financing

The illustrative budget is IDR 10 billion plus an incremental IDR 300 million annual operating allowance. The 2/3/5 billion financing split is uncommitted. Detailed costs, affordability and lifecycle funding must be established before a funding request.

7. Risks, safeguards and inclusion

We will assess service disruption, construction safety, waste/sludge handling, flood exposure, land/access, affordability and effects on vulnerable users. We will include accessible consultation, a grievance route and meaningful participation by women and underserved groups, without inventing beneficiary counts.

8. Monitoring and preparation needs

We will track service, water balance, energy, wastewater quality, reliability and distribution of benefits. Priority preparation needs are the current operating dataset, climate and hydraulic studies, contract review, design, environmental/social assessment, procurement plan and partner/funding confirmation.

Stage 4 | Financing route and report wording

Route	How we use it in this case
Utility capital and operating budgets	First check cash flow, maintenance needs, procurement authority and affordability. The reported tariff/FCR context does not prove debt capacity [1].
Local or national public support	Explore eligibility for water, sanitation and public-service priorities. The 2024 wastewater grant is historical evidence of financing, not evidence a current program is open [6].
Performance contract / private partner	Review the existing Adaro contract before funding overlapping NRW activities. New private-finance arrangements require verified savings, payment capacity and contract allocation [13].
Development finance or climate program	Consider technical assistance and a larger aggregated investment program if the pilot produces a credible pipeline. Financing terms and eligibility remain unknown.
GCF discussion route	Engage a suitable accredited entity and the Indonesian national designated authority as applicable. PT SMI is a listed accredited entity. No engagement or approval has occurred here [14, 15].

GCF's concept stage does not create a funding commitment. We would need a documented climate rationale, feasibility, environmental and social assessment, gender/inclusion work, economic and financial analysis, implementation arrangements and national coordination. For a small pilot, ordinary utility/public funding or technical assistance may be more practical; any GCF discussion should consider program scale and transaction costs. [14, 15]

Suggested paragraph for the report

We developed a worked example for Perumda Tirtawening Kota Bandung, an Indonesian utility providing water and wastewater services. We mapped published 2023 water indicators and historical Bojongsoang wastewater records to ChatWater's assessment, action-selection, project-planning and preliminary-financing stages. We explicitly labeled historical proxies, calculated values and illustrative assumptions. Because current electricity, nitrogen, sludge, hydraulic and emissions-factor data were incomplete, the example demonstrates a structured planning workflow rather than a verified utility inventory or investment appraisal. Climate-risk and water-safety inputs were prepared as supplementary notes because those application modules were not active.

Claims we can and cannot make

We can say that the guide supplies case-specific example inputs and a traceable planning rationale for all four stages. We cannot say that Tirtawening approved the case, that users completed a validated field trial, that the project is funded, or that a stated quantity of GHG emissions has been measured or reduced.

Before using the case as an actual utility submission

Information to replace or verify	Suggested source / responsible function
2023 or selected-year system input, production and billing	Utility audited water balance; production, distribution and customer-billing teams. Keep one year and one boundary.
Population and connections	Active customer and sewer registers; separate water, sewer, domestic and total accounts; document occupancy and overlap.
Energy and grid factor	PLN bills and submeter readings for each asset; current authoritative factor for the relevant grid and reporting method/year.
Wastewater volume and bypasses	Calibrated inlet/outlet meters, storm and overflow logs, pumping records and verified reuse/withdrawals.
BOD and nitrogen	Quality-assured laboratory and sampling records covering seasonal and flow variation; derive annual loads from representative data.
Sludge and methane	Sludge surveys, removal and dry-solids records; actual treatment train and suitable emission factors; biogas measurement if present.
Pump and hydraulic data	Measured flow, total dynamic head, electrical power, efficiency, duty cycle, network pressure and safe operating limits.
Climate and asset exposure	BMKG observations/projections, catchment studies, surveyed elevations, flood models and local emergency records.
Existing contracts and plans	Current Adaro scope, RPAM, business plan, RISPAM and approved capital program. Check changes since the cited records.
Costs and financing	Current engineering quantities and quotations, tax/escalation treatment, lifecycle cash flow, tariffs/charges, debt limits and funding eligibility.
Permissions and people	Asset/land rights, environmental requirements, household consent/affordability, roles, inclusion measures and grievance arrangements.

We maintain an evidence register with field name, value, unit, year, service boundary, source/page, method, evidence label, owner and date checked. If a verified figure replaces an assumption, we rerun dependent calculations and review all downstream action and funding text.

Missing inputs do not have to stop a useful discussion draft. They do stop us from representing the result as a verified emissions baseline, a final engineering design or a finance-ready proposal.

Appendix A | Historical wastewater calculation

Source: 2019 Bojongsoang study, Table 3.3 for monthly average flows and Table 3.35 for BOD. We retain the source's monthly observations and calculate totals; we do not use its inconsistent printed average row. [5]

2019 month	Inlet m^3/day	Outlet m^3/day	In BOD mg/L	Out BOD mg/L
Jan	108635	76198	23.94	13.43
Feb	109009	76691	63.32	15.03
Mar	76841	60528	85.14	36.28
Apr	82075	62546	62.19	29.52
May	82682	65011	116.25	24.41
Jun	59410	47451	219.05	15.64
Jul	55053	43970	105.44	18.69
Aug	48856	39015	86.51	23.82
Sep	35296	28176	184.52	39.36
Oct	38271	29096	150.02	19.47
Nov	52659	42122	73.19	27.22
Dec	61632	49822	69.44	25.64

Reproducible calculations

Annual flow = sum of each monthly average daily flow × days in that month. Annual BOD load = sum of each monthly flow × days × monthly BOD concentration ÷ 1,000. February 2019 has 28 days; the year has 365 days.

Derived historical quantity	Result
Inlet volume; day-weighted mean	24,566,522 m^3/year; 67,305.54 m^3/day
Outlet volume; day-weighted mean	18,829,038 m^3/year; 51,586.41 m^3/day
Influent BOD; effective flow-weighted concentration	2,250,840.14 kg/year; approximately 91.62 mg/L
Effluent BOD; effective flow-weighted concentration	438,036.28 kg/year; approximately 23.26 mg/L
Inlet minus outlet volume	5,737,484 m^3/year; unexplained in this calculation

These calculations weight monthly concentration observations by reported monthly flow. They cannot correct nonrepresentative sampling, unequal A/B coverage or inconsistent meters. The demonstration uses 25 mg/L and a closed-flow discharge assumption instead of copying this historical outlet mass; both changes are labeled A.

References | Utility and operational evidence

We checked the linked public materials for this guide. Publication years, operating years and historical proxies are kept separate. Page references refer to printed pages unless stated otherwise.

[1] Ministry of Public Works / BPKP, 2024

Buku Kinerja BUMD Air Minum 2024, Wilayah II. Tirtawening profile, p. 40 (PDF page 41), reporting 2023. Primary source for the water, financial and planning indicators. Obtained through the ministry's public performance-book distribution.

Official performance-book collection

2024 Wilayah II book

[2] BPS Kota Bandung, 2024

Statistik Daerah Kota Bandung 2024, p. 22. The 2023 volume is labeled air disalurkan. Its use as billed consumption in this guide is a proxy, not a confirmed statistical equivalence.

BPS publication page

[3] Perumda Tirtawening, service descriptions

Official, undated service pages describe raw-water sources, wastewater services, Bojongsoang and pumping stations. They establish system context, not annual operating totals.

Drinking-water service

Wastewater management

Wastewater treatment installation

[4] Perumda Tirtawening, treatment process

Proses Pengolahan Air Limbah, two-page technical description. Supports the actual pond process and preliminary treatment. The document contains a design-flow unit typo; we corroborate m^3/day with other records.

Official treatment-process PDF

[5] Tirtawening, 2019 Bojongsoang discharge study

Dokumen Kajian IPLC, government-hosted report, foreword January 2020. Tables 3.3 and 3.35, pp. 17 and 77; pond depths p. 51. Historical observations only. The indexed report was readable during research, although direct download was unavailable.

Government-hosted wastewater study

[6] Pemerintah Kota Bandung, 2024 performance report

LKjIP Kota Bandung 2024, pp. 131–134, particularly p. 134 for 577 wastewater connections and IDR 4.616 billion. Sanitation-access coverage is broader than sewer coverage.

Official city performance report

References | Methods, climate and financing

[7] WaCCliM / GIZ, ECAM Methodology Guide

Multi-stage treatment discussion p. 20; treatment equations 80–81; discharge equations 99–100; pond and nitrogen-factor tables. Method defaults are not local measurements.

ECAM Methodology Guide

[8] U.S. EPA, CREAT 3.1 Methodology Guide, 2021

Used for the manual asset, hazard, consequence and adaptation-option structure. We do not import U.S. local climate data into the Bandung case.

CREAT methodology

[9] Tirtawening, September 2, 2026

Official notice of reduced raw-water flow at Cikalong and service disruption during the dry season.

Cikalong source-flow notice

[10] Pemerintah Desa Bojongsoang, February 26, 2025

Report of local flooding and a levee breach. This is nearby hazard evidence, not a plant inundation survey.

Bojongsoang flood report

[11] BMKG, rainfall projections portal

CMIP6-based scenario framework and periods. No Bandung-specific projected percentage was extracted for this guide.

BMKG rainfall projections

[12] WHO, 2017

Climate-resilient water safety plans: Managing health risks associated with climate variability and change.

WHO climate-resilient WSP guidance

[13] Tirtawening, Adaro northern-area NRW contract

Official description of the existing performance-based arrangement. Confirm the current contract and geographic scope before investment selection.

Existing NRW contract

References | Funding process and application scope

[14] Green Climate Fund, project preparation and cycle

Current official process and preparation-resource pages. We use these for the distinction between a preliminary discussion concept and a formal, supported funding proposal. We do not claim that the illustrative project is eligible or endorsed.

Concept-note submission and screening

Proposal preparation resources

[15] Green Climate Fund, PT SMI and Indonesia

PT Sarana Multi Infrastruktur is listed as an accredited entity. It is mentioned as a possible discussion route, with no implication of engagement, financing terms or commitment.

PT SMI accredited-entity profile

GCF Indonesia country page

[16] ChatWater Resilience, live interface review

We reviewed the Stage 1 ECAM inputs, utility profile, Stage 2 custom-action form, eight Stage 3 plan fields and Stage 4 preliminary-concept inputs on October 30, 2026. The interface states that calculations use ECAM v3 formulas and identifies adaptation modules and other financing routes as planned.

ChatWater Resilience

The guide maps to the reviewed interface. Titles, available fields and generated content may change. We have not represented this document as a record of a completed, validated Bandung application run.

Plain-language glossary

Term	Meaning in this guide
PDAM / Perumda	A local public utility; the current example uses Perumda Tirtawening Kota Bandung.
IPAL / WWTP	Wastewater treatment installation / plant.
NRW	Non-revenue water: system input minus billed authorized consumption.
DMA	District metered area: a defined part of the water network with measured inflows and customer use.
BOD_5	Five-day biochemical oxygen demand, used here to describe organic pollution load.
RPAM / WSP	Rencana Pengamanan Air Minum / Water Safety Plan.
RISPAM	Master plan for the drinking-water supply system.
LLTT / IPLT	Scheduled desludging service / faecal-sludge treatment installation.
tCO_2e	Metric tonnes of carbon-dioxide equivalent, combining gases using selected warming potentials.